

**1. What is logic?**

Logic is a systematic method of reasoning that enables us to interpret the meanings of statements, examine their truth, and deduce new information from existing facts.

It plays a key role in problem-solving and decision-making.

**Note:** The study of logic begins with understanding a *statement*, which is a sentence that is either *true* or *false*, but *not both*.

**2. What is inductive reasoning (induction)?**

Inductive reasoning is when we make a general conclusion from repeated observations or experiences.

For example, a person receives a penicillin injection once or twice and experiences a reaction. He concludes that he is allergic to penicillin.

**3. What is deductive reasoning (deduction)?**

Deductive reasoning is when we draw a conclusion from already known or accepted facts.

For example, all men are mortal. We are men. So, we are mortal.

**4. What is a statement?**

A statement is a sentence or mathematical expression that is either true or false, but not both.

For example, the equation  $a = b$  is a statement, it can either be **true** or **false**, depending on the values of  $a$  and  $b$ .

**Note:** We can think of a mathematical statement as a unit of information that is either accurate or inaccurate.

**5. Give examples of true and false mathematical statements.**

**Examples of true statements:**

- (i) For a non-zero real number  $x$  and integers  $m$  and  $n$ , we have:  $x^m \times x^n = x^{m+n}$
- (ii) The sum of the measures of the interior angles of a triangle is  $180^\circ$
- (iii) The circumference of a circle with radius  $r$  is  $2\pi r$
- (iv)  $Q \subseteq R$  (The set of rational numbers is a subset of the set of real numbers)
- (v)  $\frac{22}{7} \notin Q'$  ( $\frac{22}{7}$  is not an irrational number.)
- (vi) The sum of two odd integers is always an even integer.

**Examples of false statements:**

- (i)  $3 + 4 = 8$
- (ii)  $Z \subseteq W$
- (iii) All isosceles triangles are equilateral triangles
- (iv) Between any two real numbers, there is no real number
- (v)  $\{1, 2, 3, 4\} \cap \{-1, -2, -3, -4\} = \{1, 2, 3, 4\}$
- (vi) The set of integers is finite.

**6. How do we represent statements and logical operations in symbols?**

The letters  $p, q$  etc., are used to denote statements. A brief list of commonly used logical symbols is given below:

| Symbol | How to be read | Symbolic Expression | How to be read |
|--------|----------------|---------------------|----------------|
|--------|----------------|---------------------|----------------|

|                   |                                  |                       |                                                   |
|-------------------|----------------------------------|-----------------------|---------------------------------------------------|
| $\sim$            | Not                              | $\sim p$              | Not $p$ , negation of $p$                         |
| $\wedge$          | And                              | $p \wedge q$          | $p$ and $q$                                       |
| $\vee$            | Or                               | $p \vee q$            | $p$ or $q$                                        |
| $\rightarrow$     | If... then..., implies           | $p \rightarrow q$     | If $p$ then $q$ , $p$ implies $q$                 |
| $\leftrightarrow$ | if and only if, Is equivalent to | $p \leftrightarrow q$ | $p$ if and only if $q$ , $p$ is equivalent to $q$ |

## 7. What is negation?

If  $p$  is any statement, its negation is denoted by  $\sim p$ , read as "not  $p$ ".

It follows from this definition that:

- (i) If  $p$  is true, then  $\sim p$  is false.
- (ii) If  $p$  is false, then  $\sim p$  is true.

| $p$ | $\sim p$ |
|-----|----------|
| T   | F        |
| F   | T        |

This relationship is shown in Truth Table.

## 8. What is conjunction?

The conjunction of two statements  $p$  and  $q$  is symbolically written as  $p \wedge q$  (read as " $p$  and  $q$ ").

A conjunction is considered to be true only if both statements are true.

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | F            |

This relationship is shown in Truth Table.

## 9. Determine whether the following conjunctions are true or false.

- (i) Lahore is the capital of the Punjab and Quetta is the capital of Balochistan.
- (ii)  $4 < 54 < 54 < 5$  and  $8 < 108 < 108 < 10$
- (iii)  $2+2=32 + 2 = 32+2=3$  and  $6+6=106 + 6 = 106+6=10$

**Solution:** (i) is true (both parts are true).

(ii) is true (both inequalities are true).

(iii) is false (both mathematical statements are false).

## 10. What is a disjunction?

The disjunction of statements  $p$  and  $q$  is symbolically written as  $p \vee q$  (read as " $p$  or  $q$ ").

A disjunction is true when at least one of the statements is true. It is false only when both statements are false.

This relationship is shown in Truth Table.

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| T   | T   | T          |
| T   | F   | T          |
| F   | T   | T          |
| F   | F   | F          |

## 11. Determine the truth value of the disjunction: "10 is a positive integer or 0 is a rational number."

Both statements are **true**, so the disjunction is **true**.

**12. Determine the truth value of the disjunction: "A triangle can have two right angles or Lahore is the capital of Sindh."**

Both statements are **false**, so the disjunction is **false**.

| $P$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

**13. What is an implication or conditional statement?**

A compound statement of the form "*if  $p$  then  $q$* " (symbolically written as  $p \rightarrow q$ ) is called a conditional or an implication. It may also be read as " *$p$  implies  $q$* ".

- $p$  is called the **antecedent** or **hypothesis**.
- $q$  is called the **consequent** or **conclusion**.

A conditional statement  $p \rightarrow q$  is considered to be:

- False** only when the antecedent  $p$  is **true** and the consequent  $q$  is **false**.
- True** in **all other cases**.

**14. Explain implication with a real-life situation.**

**Statement:** *If person A lives in Lahore, then he lives in Pakistan.*

Let us examine all possibilities:

- If A **does live** in Lahore and **does live** in Pakistan  $\rightarrow$  the statement is **true**.
- If A **lives in Lahore** but **not in Pakistan**  $\rightarrow$  the statement is **false** (this is the only false case).
- If A **does not live** in Lahore but **lives in Pakistan**  $\rightarrow$  we cannot say the statement is false; it's considered **true**.
- If A **does not live** in Lahore and **does not live** in Pakistan  $\rightarrow$  we still cannot reject the implication; it's regarded as **true**.

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**15. What is a biconditional or equivalence?**

The statement  $p \rightarrow q \wedge q \rightarrow p$  is shortly written as  $p \leftrightarrow q$  and is called the biconditional or equivalence. It is read as " *$p$  if and only if  $q$* ".

From the truth table, it appears that  $p \leftrightarrow q$  is true only when both statements  $p$  and  $q$  are true or both are false.

| $p$ | $q$ | $p \rightarrow q$ | $q \rightarrow p$ | $p \leftrightarrow q$ |
|-----|-----|-------------------|-------------------|-----------------------|
| T   | T   | T                 | T                 | T                     |
| T   | F   | F                 | T                 | F                     |
| F   | T   | T                 | F                 | F                     |
| F   | F   | T                 | T                 | T                     |

**16. What are the conditionals related to a given conditional statement, and how do their truth values compare?**

Let  $p$  and  $q$  be statements, and let  $p \rightarrow q$  be a given conditional. Then:

- The statement  $q \rightarrow p$  is called the **converse** of  $p \rightarrow q$ .
- The statement  $\sim p \rightarrow \sim q$  is called the **inverse** of  $p \rightarrow q$ .
- The statement  $\sim q \rightarrow \sim p$  is called the **contrapositive** of  $p \rightarrow q$ .

The truth values of the Conditional, Converse, Inverse, and Contrapositive are shown in the truth table:

| $p$ | $q$ | $\sim p$ | $\sim q$ | $p \rightarrow q$ | $q \rightarrow p$ | $\sim p \rightarrow \sim q$ | $\sim q \rightarrow \sim p$ |
|-----|-----|----------|----------|-------------------|-------------------|-----------------------------|-----------------------------|
| T   | T   | F        | F        | T                 | T                 | T                           | T                           |
| T   | F   | F        | T        | F                 | T                 | T                           | F                           |
| F   | T   | T        | F        | T                 | F                 | F                           | T                           |
| F   | F   | T        | T        | T                 | T                 | T                           | T                           |

### 17. What can be concluded from this truth table of Converse, Inverse, Contrapositive?

(i) A **conditional** and its **contrapositive** are **logically equivalent**.

→ Therefore, any theorem can be proven by proving its **contrapositive**.

(ii) The **converse** and **inverse** of a conditional are **logically equivalent** to each other.

### 18. How do you construct the truth table for $[(p \rightarrow q) \wedge p]$ and $[(p \rightarrow q) \wedge p] \rightarrow q$ ?

| $p$ | $q$ | $p \rightarrow q$ | $(p \rightarrow q) \wedge p$ | $[(p \rightarrow q) \wedge p] \rightarrow q$ |
|-----|-----|-------------------|------------------------------|----------------------------------------------|
| $T$ | $T$ | $T$               | $T$                          | $T$                                          |
| $T$ | $F$ | $F$               | $F$                          | $T$                                          |
| $F$ | $T$ | $T$               | $F$                          | $T$                                          |
| $F$ | $F$ | $T$               | $F$                          | $T$                                          |

### 19. How can we prove that the empty set $\phi$ is a subset of any set $A$ ?

Let  $U$  be the universal set. Consider the conditional:

$$\forall x \in U, \quad x \in \phi \rightarrow x \in A$$

The antecedent of this conditional is false because no  $x \in U$  is a member of  $\phi$ . Hence, the conditional is true.

### 20. What is a Mathematical Proof?

In daily life, we often need to prove our claims using solid evidence.

For example, if a student named Fayyaz comes home late, his father may doubt he attended school. Just saying "I went to school" is not enough—he needs **proof**, like:

- The **school attendance register**, or
- **CCTV footage** from the school.

Similarly, if your mobile phone breaks down and it's under warranty, the company will ask for a **warranty card** as **proof** before fixing it.

### 21. Define Mathematical Proof.

In mathematics, a *proof* is a step-by-step *logical explanation* that shows a statement is *true*. It is like *evidence* that supports a mathematical claim.

#### Note:

(i) If  $x$  is an **odd integer**, then it can be expressed in the form:

$$x = 2k + 1, \quad \text{for some } k \in \mathbb{Z}$$

(ii) If  $x$  is an **even integer**, then it can be expressed in the form:

$$x = 2k, \quad \text{for some } k \in \mathbb{Z}$$

### 22. Prove the following mathematical statements:

a) If  $x$  is an **odd integer**, then  $x^2$  is also an **odd integer**.

b) The **sum of two odd numbers is an even number**.

(a) Let  $x$  be an odd integer. By definition, an odd number can be written as:

$$x = 2k + 1, \quad \text{where } k \in \mathbb{Z}$$

Now taking square both sides:

$$x^2 = (2k + 1)^2$$

$$x^2 = (2k)^2 + 2(2k)(1) + (1)^2$$

$$x^2 = 4k^2 + 4k + 1$$

$$x^2 = 2(2k^2 + 2k) + 1$$

Let  $m = 2k^2 + 2k$ , which is an integer (since  $k \in \mathbb{Z}$ ), so:

$$x^2 = 2m + 1$$

This is the standard form of an odd number. Therefore,  $x^2$  is an odd integer, by definition.

**(b)** Let  $x$  and  $y$  be odd integers. By definition:

$$\text{and} \quad \begin{array}{ll} x = 2k + 1 & \text{where } k \in \mathbb{Z} \\ y = 2n + 1, & \text{where } n \in \mathbb{Z} \end{array}$$

Now adding both

$$\begin{aligned} x + y &= (2k + 1) + (2n + 1) \\ x + y &= 2k + 2n + 2 \\ x + y &= 2(k + n + 1) \end{aligned}$$

Let  $m = k + n + 1$ , which is an integer.

So:

$$x + y = 2m$$

This is the standard form of an even number. Therefore, the sum of two odd integers is an even integer, by definition.

**23. Prove that for any two non-empty sets  $A$  and  $B$ : (i)  $(A \cup B)' = A' \cap B'$  (ii)  $(A \cap B)' = A' \cup B'$**

**(i)  $(A \cup B)' = A' \cap B'$**

$$\begin{aligned} \text{Let } x &\in (A \cup B)' \\ \Rightarrow x &\notin (A \cup B) \\ \Rightarrow x &\notin A \text{ and } x \notin B \\ \Rightarrow x &\in A' \text{ and } x \in B' \\ \Rightarrow x &\in A' \cap B' \end{aligned}$$

Since  $x \in (A \cup B)'$  was arbitrary, we conclude:

$$(A \cup B)' \subseteq A' \cap B' \quad \dots (i)$$

$$\begin{aligned} \text{Now suppose } y &\in A' \cap B' \\ \Rightarrow y &\in A' \text{ and } y \in B' \\ \Rightarrow y &\notin A \text{ and } y \notin B \\ \Rightarrow y &\notin (A \cup B) \\ \Rightarrow y &\in (A \cup B)' \end{aligned}$$

Thus:

$$A' \cap B' \subseteq (A \cup B)' \quad \dots (ii)$$

From (i) and (ii), we conclude:

$$(A \cup B)' = A' \cap B'$$

**(ii)  $(A \cap B)' = A' \cup B'$**

$$\begin{aligned} \text{Let } x &\in (A \cap B)' \\ \Rightarrow x &\notin (A \cap B) \\ \Rightarrow x &\notin A \text{ or } x \notin B \\ \Rightarrow x &\in A' \text{ or } x \in B' \\ \Rightarrow x &\in A' \cup B' \end{aligned}$$

Since  $x \in (A \cap B)'$  was arbitrary, we conclude:

$$(A \cap B)' \subseteq A' \cup B' \quad \dots (i)$$

$$\begin{aligned} \text{Now suppose } y &\in A' \cup B' \\ \Rightarrow y &\in A' \text{ or } y \in B' \\ \Rightarrow y &\notin A \text{ or } y \notin B \\ \Rightarrow y &\notin (A \cap B) \\ \Rightarrow y &\in (A \cap B)' \end{aligned}$$

Thus:

$$A' \cup B' \subseteq (A \cap B)' \quad \dots (ii)$$

From (i) and (ii), we conclude:

$$(A \cap B)' = A' \cup B'$$

#### 24. What is a theorem in mathematics?

A theorem is a mathematical statement that has been proved to be true using logical steps based on previously accepted facts, definitions, or other theorems.

#### 25. What is the angle sum of a quadrilateral?

The sum of interior angles of any quadrilateral is  $360^\circ$ .

#### 26. What is the Fundamental Theorem of Arithmetic?

Every integer greater than 1 can be uniquely written as a product of prime numbers, ignoring the order of the factors. **OR**

Every whole number greater than 1 can be written as a product of prime numbers, in only one way (ignoring order).

#### 27. What is Fermat's Last Theorem?

There are no positive numbers  $a, b, c$  such that

$$a^n + b^n = c^n \text{ for any } n > 2.$$

This statement was made by **Pierre Fermat**, a French mathematician from the 17th century.

#### 28. What is a conjecture?

A conjecture is a mathematical statement that is believed to be true, based on observations, but not yet proven.

- If a conjecture is **proven**, it becomes a **theorem**.
- If it is **disproved**, it is considered **false**.

#### 29. What is Goldbach's Conjecture?

The **Goldbach Conjecture** says: Every even number greater than 2 is the sum of **two prime numbers**. For example,

- $4 = 2 + 2$
- $6 = 3 + 3$
- $12 = 5 + 7$

Although no one has found an even number that violates this, the conjecture has **not yet been proved**. It remains one of the oldest unsolved problems in mathematics.

#### 30. What is an axiom?

An axiom is a basic mathematical fact that is accepted as true without proof. It forms the foundation of mathematics.

For example, through a point, infinitely many lines can pass. We accept it as true based on **intuition (بصیرت)** and **experience**.

#### 30. What are some examples of axioms?

- Euclid's Axiom:** A straight line can be drawn between any two points.
- Peano's Axiom:** Every natural number has a successor.
- Axiom of Extensionality:** Two sets are equal if they have the same elements.
- Axiom of Power Set:** Every set has a set of all its subsets.

#### 31. What is the difference between an axiom and a postulate?

Both are assumed to be true without proof.

- **Axioms** apply to all branches of mathematics.
- **Postulates** are used **especially in geometry**.

### 32. What is a deductive proof?

A deductive proof is a method of proving a statement by using logical reasoning from facts that are already known to be true. For example,

**Premise (پہلو) 1:** All human beings need to breathe to live.

**Premise 2:** Ahmad is a human.

**Conclusion:** Therefore, Ahmad needs to breathe to live.

### 33. Write the converse, inverse, and contrapositive of the following conditionals: (i) $\sim p \rightarrow q$ (ii) $q \rightarrow p$ (iii) $\sim p \rightarrow \sim q$ (iv) $\sim q \rightarrow \sim p$

(i)  $\sim p \rightarrow q$

- **Converse:**  $q \rightarrow \sim p$
- **Inverse:**  $p \rightarrow \sim q$
- **Contrapositive:**  $\sim q \rightarrow p$

(ii)  $q \rightarrow p$

- **Converse:**  $p \rightarrow q$
- **Inverse:**  $\sim q \rightarrow \sim p$
- **Contrapositive:**  $\sim p \rightarrow \sim q$

(iii)  $\sim p \rightarrow \sim q$

- **Converse:**  $\sim q \rightarrow \sim p$
- **Inverse:**  $p \rightarrow q$
- **Contrapositive:**  $q \rightarrow p$

(iv)  $\sim q \rightarrow \sim p$

- **Converse:**  $\sim p \rightarrow \sim q$
- **Inverse:**  $q \rightarrow p$
- **Contrapositive:**  $p \rightarrow q$

### 34. Write the truth table of the following expressions:

(i)  $\sim (p \vee q) \vee (\sim q)$

| $p$ | $q$ | $p \vee q$ | $\sim (p \vee q)$ | $\sim q$ | $\sim (p \vee q) \vee (\sim q)$ |
|-----|-----|------------|-------------------|----------|---------------------------------|
| $T$ | $T$ | $T$        | $F$               | $F$      | $F$                             |
| $T$ | $F$ | $T$        | $F$               | $T$      | $T$                             |
| $F$ | $T$ | $T$        | $F$               | $F$      | $F$                             |
| $F$ | $F$ | $F$        | $T$               | $T$      | $T$                             |

(ii)  $\sim (\sim q \vee \sim p)$

| $p$ | $q$ | $\sim p$ | $\sim q$ | $\sim q \vee \sim p$ | $\sim (\sim q \vee \sim p)$ |
|-----|-----|----------|----------|----------------------|-----------------------------|
| $T$ | $T$ | $F$      | $F$      | $F$                  | $T$                         |
| $T$ | $F$ | $F$      | $T$      | $T$                  | $F$                         |
| $F$ | $T$ | $T$      | $F$      | $T$                  | $F$                         |
| $F$ | $F$ | $T$      | $T$      | $T$                  | $F$                         |

(iii)  $(p \vee q) \leftrightarrow (p \wedge q)$

| $p$ | $q$ | $p \vee q$ | $p \wedge q$ | $(p \vee q) \leftrightarrow (p \wedge q)$ |
|-----|-----|------------|--------------|-------------------------------------------|
| $T$ | $T$ | $T$        | $T$          | $T$                                       |
| $T$ | $F$ | $T$        | $F$          | $F$                                       |
| $F$ | $T$ | $T$        | $F$          | $F$                                       |
| $F$ | $F$ | $F$        | $F$          | $T$                                       |

### 35. Differentiate between a mathematical statement and its proof. Give two examples.

| Mathematical Statement                                                                                                 | Mathematical Proof                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A mathematical statement is a sentence or mathematical expression that is either true or false but not both.           | A proof is a logical explanation that verifies a statement using definitions, axioms, known theorems, or logical reasoning.                                                                 |
| (i) If $x$ is an odd integer, then $x^2$ is also an odd integer.<br>(ii) The sum of two odd numbers is an even number. | (i) If $x = 2k + 1 \Rightarrow x^2 = 2m + 1 \rightarrow$ Odd (proved using algebra).<br>(ii) If $x = 2k + 1, y = 2n + 1 \Rightarrow x + y = 2m \rightarrow$ Even (proved using identities). |

### 36. Difference between an Axiom and a Theorem.

| Axiom                                                                                                                                                         | Theorem                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An axiom is a mathematical statement accepted without proof. It is based on basic facts or everyday experience and forms the foundation of further reasoning. | A theorem is a mathematical statement that has been proven true using axioms, previously known theorems, and logical steps.                                                                         |
| (i) Through a given point, infinitely many lines can pass.<br>(ii) <b>Euclid's Axiom:</b> A straight line can be drawn between any two points.                | (i) The sum of interior angles of a quadrilateral is $360^\circ$ .<br>(ii) <b>Fundamental Theorem of Arithmetic:</b> Every integer greater than 1 can be expressed uniquely as a product of primes. |

### 37. What is the importance of logical reasoning in mathematical proofs? Give an example to illustrate your point.

**Logical reasoning** is essential in mathematics because it allows us to **prove or disprove** statements using facts, definitions, and structured thinking. Without logical reasoning, we cannot be certain whether a mathematical statement is valid.

Consider this deductive reasoning:

- **Premise 1:** All human beings need to breathe to live.
- **Premise 2:** Ahmad is a human.
- **Conclusion:** Therefore, Ahmad needs to breathe to live.

This same type of reasoning is used in **mathematical deductive proofs**. For example, in algebra, we use identities and rules to show that both sides of an equation are equal through step-by-step logic. This ensures the result is always true if the premises are true.

### 38. Identify: Axiom, Conjecture, or Theorem. (i) There is exactly one straight line through any two points. (ii) Every even number greater than 2 can be written as the sum of two primes. (iii) The sum of the angles in a triangle is $180^\circ$

| Statement                                                                      | Type       | Reasoning                                                                             |
|--------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------|
| (i) There is exactly one straight line through any two points.                 | Axiom      | This is a basic assumption in Euclidean geometry, accepted without proof.             |
| (ii) Every even number greater than 2 can be written as the sum of two primes. | Conjecture | This is the Goldbach Conjecture, believed to be true but not yet proven.              |
| (iii) The sum of the angles in a triangle is $180^\circ$                       | Theorem    | This is a proven result in Euclidean geometry, based on axioms and logical reasoning. |